CIE

COPYRIGHT INFORMATION EXTRACTION FORM

TITLE: **Career Readiness and Personalized Roadmap System Using Machine Learning**

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

Name: **Kshitij Raj Shukla**

Roll No/ Employee Code: **2210990526**

All MEMBERS DETAILS INCLUDING SUPERVISOR

NAME	EMP CODE/ROLL	ADDRESS WITH EMAIL AND MOBILE NO.
Kshitij Raj Shukla	2210990526	kshitij526.be22@chitkara.edu.in ,8558090489
Mayank Kumar	2210990568	mayank568.be22@chitkara.edu.in ,8168887165
Mayank Bisht	2210990566	mayank566.be22@chitkara.edu.in ,8968782095
Dr. Gifty Gupta		gifty.gupta@chitkara.edu.in, 9996303910

ABSTRACT

1.1 Problem Background

- Navigating modern career landscapes is increasingly complex due to **rapid technological advancements** and evolving job market requirements.
- Individuals often struggle to align their **current skills, educational backgrounds, and personal interests** with optimal career paths.
- Traditional career counselling is **limited in scalability**, lacks **data-driven insights**, and fails to provide **actionable, personalized guidance**.
- This leads to **skill gaps, underemployment, and decision paralysis** among students and professionals.
- Therefore, there is a critical need for an **automated, intelligent system** that can assess individual profiles and generate **personalized career roadmaps with skill gap analysis**.

1.2 Objective of the Project

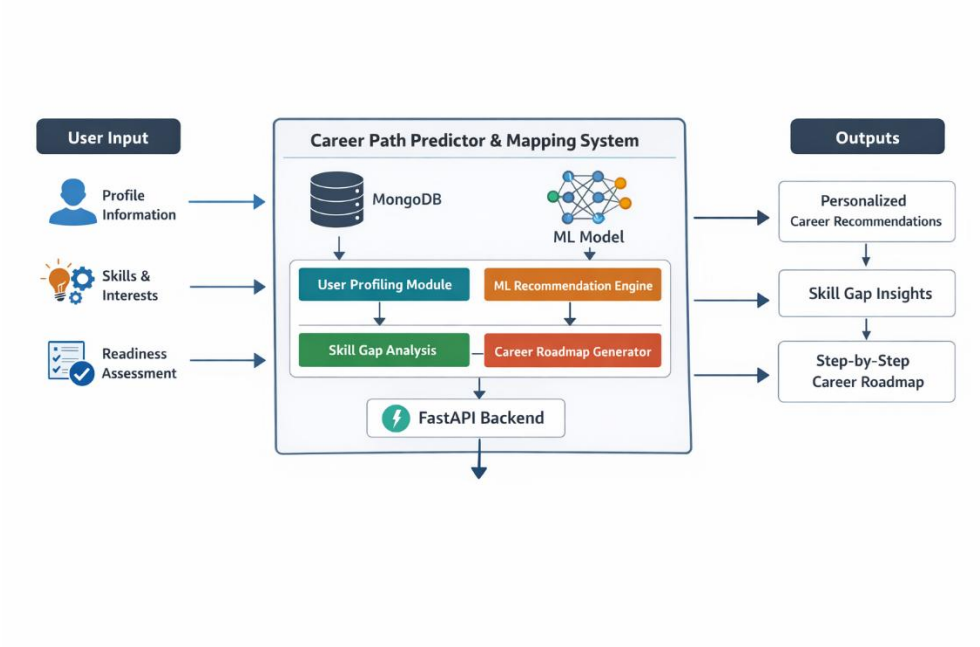
- The primary objective is to develop an **intelligent web-based platform** that leverages **machine learning** to provide **personalized career recommendations**.
- The system aims to:
  - Assess **user readiness**
  - Identify **skill gaps**
  - Generate **structured, actionable career roadmaps**.

1.3 Proposed Solution / System Overview

- The proposed system, **"Career Path Predictor & Mapping System"**, is a **comprehensive AI-driven web application** for automated career guidance.
- It evaluates **user profiles, interests, and competencies** against **industry requirements** using **predictive machine learning models**.
- The system consists of:
  - **User Profiling Module**
  - **Machine Learning Recommendation Engine**

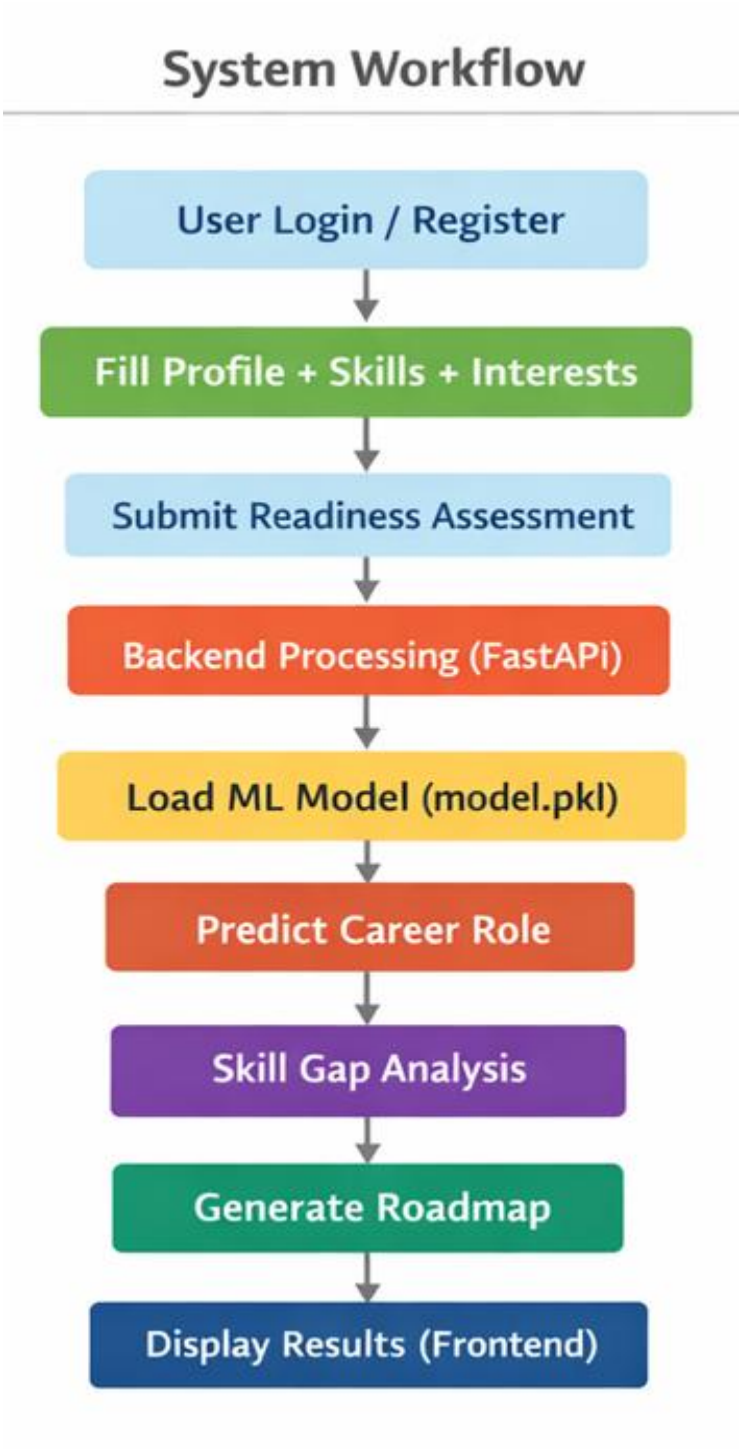
- **Readiness Assessment Interface**
- **Career Roadmap Generation Service**

- These components work together to:
  - Provide **personalized insights**
  - Enable **continuous progress tracking**
  - Clearly define **skill acquisition paths**



1.4 **Key Features or Functionality**

- **Centralized Dashboard** for tracking progress
- **Secure User Authentication System**
- **Secure User Authentication System**
- **Readiness Assessment System** with dynamic surveys to evaluate skills
- **Skill Gap Identification Engine**
- **Career Roadmap Generator** that provides:
  - Step-by-step milestones
  - Recommended **topics, certifications, and projects**



1.5 **Technology / Method Used**

- The project utilizes a decoupled architecture.
  - The frontend is built with:
    - **Next.js (React)**
    - **TypeScript**
    - **Tailwind CSS**

- RESTful backend:
    - **FastAPI (Python)**
    - **MongoDB** for data management

- The recommendation and assessment engines (machine learning models)
    - **Scikit-Learn**
    - **Pandas**
    - **NumPy**
    - **Random Forest Algorithms (Classification and Regression):**
        - Career prediction
        - Skill gap analysis

1.6 Expected Outcome & Benefits

- Reduces **decision fatigue** in career planning
- Provides **clear and personalized career direction**
- Accelerates **skill acquisition and learning efficiency**
- Improves **employability and job readiness**
- Acts as a **scalable, data-driven career guidance platform**

1.7 Original Contribution Statement

- This project introduces a **novel integration of machine learning-based career prediction with dynamic skill gap analysis.**
- It uniquely transforms career guidance into **fully personalized, step-by-step actionable roadmaps**, bridging the gap between **career planning and execution.**

1.8 Reference

- The following resources, libraries, and frameworks were used during the development of this project:
    - **FastAPI Documentation –** https://fastapi.tiangolo.com/
    - **MongoDB Documentation –** https://www.mongodb.com/docs/
    - **Scikit-learn Documentation –** https://scikit-learn.org/
    - **Joblib Documentation –** https://joblib.readthedocs.io/
    - **Pydantic Documentation –** https://docs.pydantic.dev/
    - **Python Official Documentation –** https://docs.python.org/3/
- **All source code, system design, and implementation are original and developed by the applicant. External resources were used only for reference and learning purposes.**